

H2-SHIFT Round 2: judge Q&A, validation log and change log

Team Wookies | Raunak Jain, Abhishek Aby, Divya | NIT Warangal | Research cut-off 14 September 2026

1. Judge Q&A (20 strongest questions)

1. What is distinctive about the proposal?

It converts the installed factory-CNG base rather than building new vehicles, keeps petrol as an automatic backup so no car is stranded by missing stations, and, unlike most hydrogen pitches, states the price at which it works (₹188/kg delivered for CNG fuel parity) and the gates at which it stops.

2. Why retrofit this vehicle?

The Dzire Tour S CNG is the fleet workhorse with a 4-cylinder 1.2 L engine (power margin after the hydrogen derate), a boot bay that already carries a 55 L cylinder, hardened valve seats, a 37 L petrol tank for backup, and the largest donor pool (Maruti sells about 70 percent of India's CNG cars).

3. Why would the customer choose it over staying on CNG or buying a BEV?

On money alone they would not: continuing on CNG costs ₹5.2/km, a compact BEV ₹5.6/km including purchase, and the retrofit ₹13.7/km at 2028 hydrogen prices. The customer for the pilot is a fleet with a decarbonisation mandate or a public programme paying for abatement and data; the commercial customer appears only if delivered hydrogen falls below about ₹190/kg.

4. Does the hydrogen storage system actually fit?

A 60 L composite cylinder (OD 400 mm, 767 mm long, 36 kg) fits the transverse bay used by the 55 L CNG cylinder (OD 316 mm, about 900 mm), but it is 84 mm larger in diameter, so the boot loses a further 40 to 60 L. No under-floor option exists at 163 mm ground clearance. Phase 0 includes a physical mock-up because this is a concept, not validated design work.

5. What is sacrificed in passenger capacity, luggage or payload?

Seating is unchanged (five). Luggage falls from roughly 200 L with CNG to about 150 L. Kerb mass falls by about 13 kg, so payload is unchanged or slightly better. Peak power in hydrogen mode is about 42 PS, 48 percent below petrol; the car switches to petrol automatically for high load.

6. Which components cannot be reused?

Every CNG fuel-system part: the steel cylinder (ISO 11439 steels are limited to 2 percent hydrogen), regulator, rail, injectors, hoses and lines. Hydrogen parts need UN R134 or ISO 19881 class qualification. The engine, gearbox, petrol system, catalyst and body are retained.

7. What is the credible pilot conversion cost?

₹7.7 lakh per car including GST, dominated by the imported cylinder and valve at small quantities; ₹3.2 lakh installed at 10,000 kits per year (₹2.3 lakh ex-works). Round 1's ₹1.8 to 2.2 lakh is roughly the ex-works volume figure without taxes and margin.

8. Where will hydrogen come from and at what delivered price?

Certified green hydrogen by tube trailer from a SIGHT-awarded producer to a shared dispensing point in Faridabad. Our build-up gives about ₹1,433/kg at the pilot point (150 kg/day, 50 percent used) and about ₹584/kg at a 2030 hub (500 kg/day, 70 percent used); CEEW's central 2030 estimate is ₹360/kg. India's record refinery-gate tender is ₹279/kg before compression and dispensing.

9. What if fuel cost stays higher than expected?

The pilot still delivers its purpose (approval route, real-world data, supplier readiness) because fuel is a programme cost. The scale decision at gate 4 simply fails, and the programme stops and publishes.

10. What if the station is unavailable?

Every car runs on petrol automatically (555 km on a tank); the fleet loses the hydrogen benefit, not the service. The station partner carries utilisation risk through a take-or-pay contract, and a 1.5-day cascade buffer covers delivery slips.

11. What is the exact approval uncertainty?

There is no standard or rule for a hydrogen retrofit kit. AIS-195 covers new hydrogen ICE vehicles; AIS-157 (fuel cell) explicitly excludes retrofits; Rule 126 and Rule 115 provide the retrofit machinery only for CNG and LPG. We need either an AIS-195 extension or a Rule 115 hydrogen-kit provision; the trial itself can run under a trade certificate with test-agency evaluation, as the Mirai and Tata trials did.

12. How is NOx addressed?

Lean burn above lambda 2 gives near-zero engine-out NOx at steady state; transients do not stay lean, so the kit carries a lean NOx trap provision and lambda control, and the gate-2 test requires 60 mg/km or better on the BS-VI cycle. This is a target to demonstrate.

13. What is the environmental benefit under matched assumptions?

With certified green hydrogen (2 kg CO₂e/kg) the converted car emits about 89 g CO₂e/km well-to-wheel including the kit, against 137 on CNG and 134 for a new BEV on the 2030 grid including its manufacturing. With grid-electrolysis hydrogen it would be four times worse than CNG, so the fuel's certification is a condition, not a detail. Hydrogen combustion also emits NOx.

14. Who funds development and underutilised infrastructure?

Proposed split of ₹37.6 crore: Suzuki R&D Center India and Maruti 40 percent (engineering, prototypes, test cells), public support 35 percent (the MNRE transport pilot scheme names 4-wheelers; ARAI runs its calls), an oil marketing company 15 percent (dispensing point), the fleet 10 percent. Station utilisation risk sits with the station partner under take-or-pay. None of this is committed.

15. What result would make you stop?

No written regulator route by March 2027; backfire or NOx not controlled within 500 test-cell hours; delivered hydrogen still above ₹250/kg in 2030 with no credible path to ₹190/kg; or fleets refusing the power and boot loss in Phase 0 interviews.

16. What can realistically be achieved by 2030?

An approved retrofit route, 30 cars with 18 months of independent data, a shared hydrogen dispensing point in Faridabad, supplier quotations for car-size cylinders, and a decision. Conditional expansion to 1,000 to 1,500 cars at up to 8 hub stations is possible only after gate 4; Round 1's 1,00,000 cars is not.

17. Why not a fuel-cell conversion for better efficiency?

It halves hydrogen use per km but replaces the entire powertrain (stack, battery, motor, inverter, cooling, tank, controls) at ₹15 to 25 lakh per car in a pilot, more than a new BEV. It is not a retrofit in cost terms and has no retrofit approval route either.

18. Why keep the HCNG step if it displaces only 6 percent of energy?

Because it is legal today (GSR 585(E), IS 17314), needs no vehicle change, and lowers CO and HC; it is an air-quality measure that lets CGD companies start blending. We no longer present it as decarbonisation.

19. How do you know fleets want this?

We do not yet. Customer demand is a stated hypothesis; Phase 0 funds 30 structured interviews and telematics on 20 donor cars, and gate 1 requires evidence from at least three fleets before any engineering spend.

20. Is Suzuki endorsing this?

No. Suzuki R&D Center India is proposed as engineering lead because of its CNG engine and calibration know-how and Suzuki's own hydrogen combustion prototypes (Burgman, Swift with AVL). No endorsement or partnership is claimed.

2. Validation log

Item	What was done
Attachments	Both PDFs rendered and inspected page by page (brief: 3 pages; submission: 4 slides at 1440 x 810 pt). Title slide preserved as the native 1376 x 768 raster with the source's 3.14 pt white bands; pixel comparison of renders confirms visual identity.
Research	Four workstreams (engineering; safety and regulation; market, infrastructure and benchmarks; economics and emissions) plus a gap-closure pass. Research cut-off 14 September 2026. 309 register entries.
Access limitation	The build environment's network policy blocked direct access to most primary hosts (ARAI, MoRTH, PIB, PESO, MNRE, DOE, NIST, IEA, OEM sites). Findings rely on indexed excerpts of those pages; PIB releases of 14 Feb 2024 and 3 Mar 2025 and the Scribd copy of AIS-157 were read in full through the search index. Entries marked 'excerpt' or 'summary' should be re-verified from the originals before external use.
Model	Workbook built with formulas only, recalculated with LibreOffice Calc; zero error cells; decomposition checks (a + b km + c km P) reproduce the TCO differences to zero; scenario selector tested for 1, 2, 3.
Consistency	Slides, annexure and pitch read numbers from the same output file exported from the recalculated workbook; budgets reconcile to the funding ask; rollout quantities reconcile to station capacity.

Design	Every slide rendered and inspected; overlaps and clipping fixed; fonts limited to Arial for cross-platform fidelity; PPTX and PDF each below 2 MB.
Not done	No customer interviews, letters of interest, test results, supplier quotations or partnerships exist; all are labelled as hypotheses or proposals. The submission portal's annexure handling could not be verified (site inaccessible).

3. Change log: Round 1 to Round 2

Element	Round 1	Round 2
Core idea	Convert the installed CNG car base to hydrogen, fleets first, aggregate demand at hubs	Retained
Fallback fuel	CNG retained with automatic H2/CNG switching (second cylinder)	Petrol retained (already in every factory CNG car); CNG hardware removed; two cylinders do not fit
Donor	CNG cars in general (WagonR image)	Dzire Tour S CNG primary; WagonR, Ertiga, Eco assessed as alternatives
Retrofit ladder	HCNG (2026-27), bi-fuel H2 plus CNG (2027-28), dedicated H2-ICE (2029-30)	HCNG as an air-quality step (about 6 percent of energy), H2-ICE retrofit pilot (2027-30), conditional 700 bar or hub-scale option after 2030
Hydrogen price basis	Production-cost trajectory to ₹160-200/kg treated as pump price	Delivered price build-up: ₹1,433/kg pilot, ₹584/kg 2030 hub (our conservative), ₹360/kg CEEW central
Economics	₹1.5 lakh annual saving versus petrol, 18-month payback	Fair three-way TCO (continue CNG, retrofit, sell and buy BEV): retrofit never pays back within 5 years; parity conditions stated
Kit cost	₹1.8 to 2.2 lakh	₹7.7 lakh pilot; ₹3.2 lakh installed at volume (₹2.3 lakh ex-works)
Range and power	400 plus km combined	61 km per hydrogen fill plus 555 km petrol; 42 PS in hydrogen mode
Regulation	AIS-157 plus PESO plus ISO/SAE	AIS-195 (new H2-ICE vehicles), Rule 126/115 retrofit template, PESO GCR 2025, Form-H; no retrofit route exists; gate 1 added
Roadmap	500 cars and 100 stations (2026-27), 25,000 kits (2027-28), 1,00,000 plus cars (2029-30)	3 prototypes (2027-28), 30 pilot cars (2028-30), conditional 1,000 to 1,500 cars after 2030; four gates
Impact	1,00,000 cars, 0.9 Mt CO ₂ , 15,000 t/yr hydrogen, ₹2,000 crore industry, 10,000 jobs	About 1,400 cars, 3,600 t CO ₂ e/yr, 924 t/yr hydrogen by 2030; jobs not claimed
Market figures	7.5 million CNG vehicles, 65 percent cars, 4,000 retrofit centres	8.2 million (Mar 2025); car share estimated 55 percent; retrofit-centre count unverified
Benchmarks	Toyota GR Corolla and Hilux H2-ICE	HiAce (Australia) is the hydrogen combustion van; Hilux was a fuel-cell prototype; Suzuki-AVL Swift added; global setbacks added
Carbon credits	USD 10 to 15/t as revenue	Immaterial; treated as upside only
Claims removed	Job creation, 'zero range anxiety' as absolute, 'CNG cylinder retained', implied component reuse, 'BS-VI compliance pathway' as achieved	Replaced by evidence-labelled statements

4. Internal review framework (our own, not the organiser's scoring)

We reviewed the deck as an automotive R&D engineer (physics, packaging, power, NO_x), a fleet operator (downtime, boot, refuelling, cost per km), a type-approval specialist (route, standards, evidence), a finance reviewer (fair TCO, no double counting, discounting) and a competition judge (answers the brief, builds on Round 1, honest). Corrections made during review: fallback fuel changed to petrol; kit cost raised; pump-price basis; payback claim withdrawn; roadmap and impact rebased; AIS-157 versus AIS-195 corrected; HiAce versus Hilux corrected; job claim removed. Official scoring weights were not published to us and are not assumed.

The Round 1 deck lists Raunak Jain first but does not name a team leader. The submission filename stem Raunak_TeamWookies_NITWarangal assumes Raunak Jain is the leader; confirm before uploading.